

Enhancing Fraud Detection Using Synthetic Transactions Generated by GenAI

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EXECUTIVE SUMMARY

Credit card fraud is a growing problem worldwide, causing significant financial losses for both consumers and financial institutions. This project aims to build a smart system that can automatically detect fraudulent credit card transactions by analyzing patterns in transaction data.

Using historical transaction data from real credit card users, the system learns to distinguish between normal and suspicious activities. It looks for unusual spending behavior, such as abrupt changes in transaction amount or timing, to flag potential fraud in real time. But the problem is that there is low fraud detection recall due to class imbalance. To overcome this, synthetic samples are generated using GAN model.

This proactive fraud detection helps protect cardholders from unauthorized charges, reduces losses for banks, and improves overall trust in digital payments. The solution is implemented as an easy-to-use web application, enabling quick and convenient fraud checks with minimal manual effort by training the model on synthetic samples generated.

Ultimately, this project leverages advanced machine learning techniques to improve overall performance by increasing ROC-AUC score of preprocessed data by 7% and recall by 4% and it enhances security in the credit card ecosystem, providing a safer experience for consumers and businesses alike.

PROJECT OBJECTIVES

I. Improving Fraud Detection

Credit card fraud detection is challenging due to the rarity of fraudulent transactions compared to normal ones, making it difficult for machine learning models to learn distinguishing patterns effectively. Improving fraud detection means developing techniques that can better identify fraudulent transactions while minimizing false alarms and missed fraud cases.

II. Addressing Class Imbalance with GenAI Data Augmentation

The dataset typically has a severe class imbalance, with fraudulent transactions being a small minority. This imbalance makes it hard for models to detect fraud because they mostly see legitimate transactions during training.

To solve this, Generative AI (GenAI) models—such as GANs (Generative Adversarial Networks) or other synthetic data generators—can create realistic synthetic samples that mimic the minority class (fraud cases). Augmenting real data with these synthetic samples:

- ✓ Balances the dataset without discarding any legitimate data.
- ✓ Preserves data privacy since synthetic data does not contain actual user information.
- ✓ Helps models learn more diverse fraud patterns, improving generalization.

III. Preserving Data Privacy

Generating synthetic fraud data helps avoid exposing real sensitive financial information. These synthetic samples can be shared and used for training without risking privacy breaches. This is crucial in regulatory environments where data privacy must be maintained.

IV. Improving Machine Learning Models by Generating More Samples

More fraud samples, especially synthetic ones, allow the model to:

- ✓ Learn rare but important fraud patterns that may be missed in imbalanced datasets.
- ✓ Reduce overfitting to limited fraud examples.
- ✓ Improve robustness and accuracy.

By enhancing the data diversity and quantity, the ML model can better differentiate genuine transactions from fraud.

V. Enhancing Recall without Sacrificing Precision or Introducing Bias

Recall measures how many actual frauds are correctly detected. Increasing recall reduces missed frauds but can sometimes increase false alarms (lower precision).

Augmenting minority class samples carefully with GenAI-generated data helps improve recall while:

- ✓ Maintaining precision by providing high-quality, representative samples.
- ✓ Avoiding bias by ensuring synthetic data covers the variety of fraud cases without overrepresenting any subtype.
- ✓ Using evaluation metrics such as precision-recall curves and AUC rather than plain accuracy ensures a balanced improvement.
- ✓ Proper validation and hyperparameter tuning help keep the balance.

DATASET OVERVIEW

Dataset Source: Kaggle - Credit Card Fraud Dataset

Format: The dataset contains 284,807 credit card transactions recorded over two days.

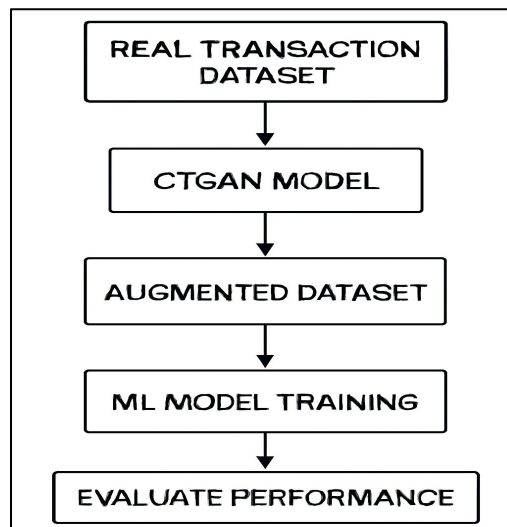
Each transaction has 31 features:

- ✓ **Time:** Number of seconds elapsed between this transaction and the first transaction in the dataset.
- ✓ **V1 to V28:** Anonymized principal components obtained through PCA for privacy reasons (to mask sensitive information).
- ✓ **Amount:** The transaction amount.
- ✓ **Class:** The target variable, where 0 represents a legitimate transaction and 1 indicates fraud.

Key Characteristics:

- ✓ The dataset is highly imbalanced, with only 492 fraud cases (about 0.17% of all transactions), which presents a challenge for modeling.
- ✓ Fraudulent transactions tend to have higher average transaction amounts compared to legitimate ones.
- ✓ Features V1 through V28 are the result of dimensionality reduction, designed to protect privacy while preserving important variance in the data.

TECHNICAL ARCHITECTURE



The flowchart illustrates a workflow for fraud detection where:

- ✓ A real transaction dataset is first used to filter out fraud samples.
- ✓ These fraud samples are used to train a CTGAN model (Conditional Tabular GAN).
- ✓ The CTGAN model then generates 5,000 synthetic fraud samples to augment the original dataset.
- ✓ This augmented dataset (real + synthetic data) is used for training a machine learning model.
- ✓ Finally, the trained model's performance is evaluated.

This process is focused on improving fraud detection by augmenting imbalanced real data with synthetically generated fraudulent transactions using CTGAN.

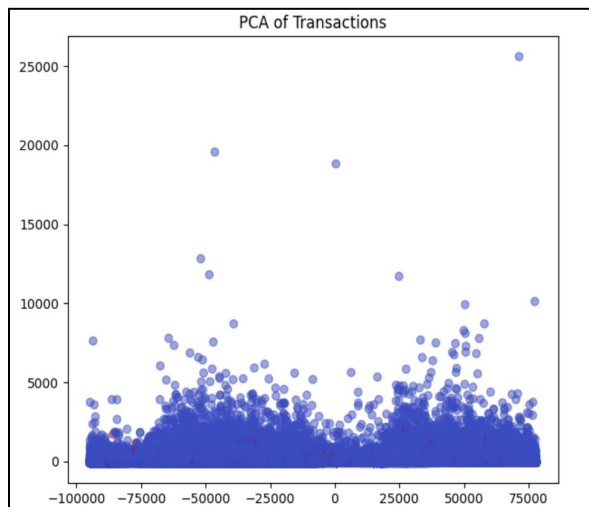
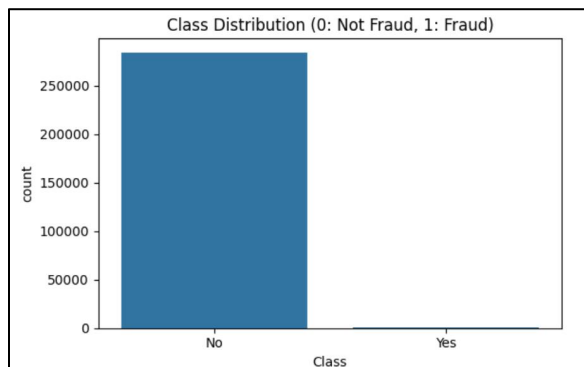
METHODOOGY

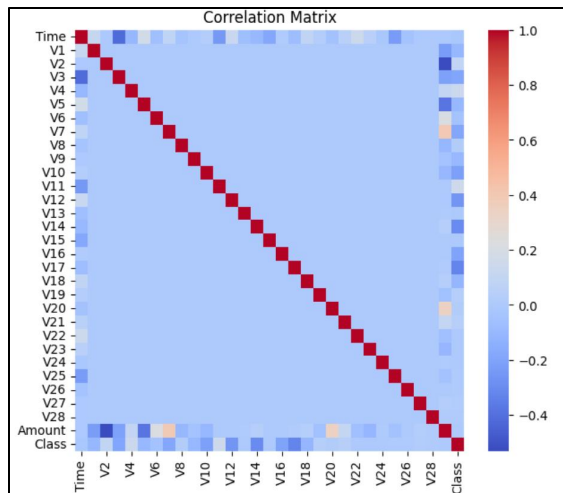
1. Exploratory Data Analysis (EDA) and Data Preprocessing

Data Understanding:

- ✓ Load the real transactions dataset.
- ✓ Explore dataset shape, feature types, and basic statistics.
- ✓ Visualize distributions of key features such as transaction amounts, transaction types, and time.
- ✓ Identify class imbalance between fraudulent and legitimate transactions.
- ✓ Examine correlations between features to identify important predictors.

Here are the plots to give EDA insights:





Data Cleaning:

- ✓ Handle missing values using imputation (mean/median/mode/interpolation).
- ✓ Remove duplicate transactions to ensure data integrity.
- ✓ Detect and treat outliers that may distort model training.

Feature Engineering:

- ✓ Select relevant features based on domain knowledge and correlation analysis.
- ✓ Encode categorical variables using techniques like one-hot encoding.
- ✓ Normalize or scale numerical features to ensure uniform contribution to the model.

2. CTGAN Model for Synthetic Data Generation

Train CTGAN Model:

- ✓ Use the filtered fraud transactions from the real dataset to train a Conditional Tabular GAN (CTGAN).
- ✓ CTGAN learns the distribution of fraud samples, preserving complex relationships between features.

- ✓ Take training epochs = (50, 100, 200) to run CTGAN model.

Generate Synthetic Fraud Samples

- ✓ Use the trained CTGAN to generate a large number of synthetic fraud examples (e.g., 5,000).
- ✓ Validate synthetic data quality by comparing distributions and feature correlations with real fraud data.

Augment Dataset:

- ✓ Combine synthetic fraud samples with the original real transaction dataset.
- ✓ This enriched dataset better balances classes and helps the model detect rare fraud patterns.

3. Machine Learning Model Training and Evaluation

Model Selection and Training:

- ✓ Choose appropriate classifiers like Logistic Regression, Random Forest or XGBoost which handle tabular data well.
- ✓ Train the model on the augmented dataset and real data as well.

Performance Evaluation

- ✓ Evaluate the model using metrics suited for imbalanced classification such as:
 - Precision, Recall, F1-score (focus on recall to catch frauds)
 - ROC-AUC curve
 - Confusion matrix for detailed error analysis
- ✓ Compare the performance of models trained with and without synthetic data augmentation.

MODEL RESULTS & ANALYSIS

1. Comparison of Performance of Different Algorithms:

	Model	ROC-AUC (Original)	ROC-AUC (Augmented)	F1-Score (Original)	F1-Score (Augmented)	Recall (Original)	Recall (Augmented)	Precision (Original)	Precision (Augmented)
0	Logistic Regression	0.955981	0.777248	0.716535	0.438292	0.614865	0.283374	0.858491	0.966874
1	Random Forest	0.930739	0.990540	0.845283	0.639706	0.756757	0.475121	0.957265	0.978750
2	XGBoost	0.928599	0.992197	0.838951	0.870239	0.756757	0.783374	0.941176	0.978772

From the given table it can be concluded that XGBoost performed well in case of both augmented data and preprocessed original data in every metric. And it performed much better in case of augmented data.

2. Metrics Comparison on Different Datasets:

	Model Dataset	F1-Score	Recall	Precision	ROC-AUC
0	Real	0.838951	0.756757	0.941176	0.928599
1	Real + Synthetic	0.870239	0.783374	0.978772	0.992197

This table shows how adding synthetic samples to the dataset improved the overall performance of the trained model.

(Note: For visual plots, check Appendix)

BUSINESS IMPACT

1. Higher Catch Rate of Rare Fraud Events:

- ✓ Fraud instances are scarce and diverse, causing imbalanced datasets that limit model learning.
- ✓ CTGAN generates realistic synthetic fraud samples that represent rare and complex fraud patterns.
- ✓ This increased diversity of fraud data improves the model's ability to detect rare fraud events, boosting the catch rate and reducing false negatives.
- ✓ Better detection of rare fraud cases protects the business from significant financial losses.

2. Improved Performance of the Model:

- ✓ The inclusion of CTGAN synthetic data effectively augments the training dataset, mitigating class imbalance.
- ✓ Models trained on this augmented dataset show higher recall, precision, and overall accuracy.
- ✓ Enhanced ability to capture complex, non-linear relationships and edge cases leads to more robust fraud prediction.
- ✓ Improved performance metrics such as ROC-AUC and PR-AUC translate directly into more reliable fraud detection decisions.

3. Reduced Fraud Leakage through Better Model Generalization:

- ✓ Real fraud data is often limited, increasing the risk of overfitting to known fraud types.

- ✓ Synthetic samples generated by CTGAN expand the representation of fraud types, enabling the model to generalize better to unseen and novel fraud patterns.
- ✓ This reduces fraud leakage, where fraudulent transactions go undetected, maintaining business reputation and reducing losses.

4. Synthetic Generation Avoids Using Real Customer Data:

- ✓ Synthetic fraud data is generated to mimic the patterns of real data without including any actual personal or sensitive customer information.
- ✓ This helps maintain strict data privacy and compliance with regulations like GDPR and CCPA.
- ✓ Organizations can safely share synthetic datasets for research, collaboration, or model improvement without risking customer data exposure.

5. Training Data Can Be Increased Artificially as Needed:

- ✓ CTGAN allows on-demand generation of synthetic fraud data, enabling dynamic adjustment of dataset size.
- ✓ This flexibility means training datasets can be scaled up to meet model complexity requirements or evolving fraud scenarios.
- ✓ Artificially increasing fraud samples helps maintain model efficacy over time despite changing fraud tactics.

RISK & LIMITATIONS

Risks/Challenges:

1. Training Instability in CTGAN:

- ✓ GAN models, including CTGAN, can suffer from unstable training, such as mode collapse where synthetic data lacks diversity.
- ✓ This results in synthetic samples that do not fully represent the range of fraud behaviors.

2. Fidelity vs Privacy Trade-Off:

- ✓ Synthetic data aims to resemble real data well, but overly realistic samples risk inadvertently revealing sensitive information.
- ✓ Ensuring privacy while maintaining data utility is challenging.

3. Computational Resource Demands:

- ✓ Training deep generative models like CTGAN requires significant computational power (GPUs) and time.
- ✓ Resource constraints can hinder timely model development.

4. Synthetic Data Quality and Bias:

- ✓ Poor quality synthetic data may introduce biases or artifacts that mislead the fraud detection model.
- ✓ If the original fraud data is very limited or biased, CTGAN will replicate these issues.

5. Model Overfitting on Synthetic Samples:

- ✓ Excessive reliance on synthetic data during training can cause models to overfit on synthetic patterns that do not generalize well to real transactions.

6. Explainability and Transparency Challenges:

- ✓ GANs are black-box models, making it difficult to understand how synthetic samples are generated.
- ✓ This opacity can reduce trust among stakeholders and complicate regulatory compliance.

7. Class Boundary Distortion:

- ✓ Synthetic data may blur the decision boundary between fraud and non-fraud, possibly deteriorating classifier performance on majority class transactions.

Mitigation Strategies:

1. Careful Hyperparameter Tuning & Monitoring:

- ✓ Use advanced GAN training techniques and regular monitoring to avoid mode collapse.
- ✓ Validate synthetic data diversity throughout training.

2. Privacy-Preserving Techniques:

- ✓ Incorporate differential privacy or data anonymization safeguards.
- ✓ Regularly audit synthetic data to ensure no real records are exposed.

3. Leverage Cloud and Scalable Resources:

- ✓ Use cloud-based GPUs or distributed training to meet computational requirements efficiently.

4. Quality Assessment and Filtering:

- ✓ Conduct statistical and visual analysis comparing synthetic and real data distributions.
- ✓ Filter out low-quality or unrealistic synthetic samples before augmenting data.

5. Balanced Use of Synthetic and Real Data:

- ✓ Mix synthetic fraud samples carefully with real data.
- ✓ Apply robust validation and cross-validation to prevent overfitting.

6. Incorporate Explainable AI Techniques:

- ✓ Use explainability tools to interpret downstream fraud detection models.
- ✓ Develop documentation and transparency reports on synthetic data generation.

7. Boundary-aware Synthetic Sampling:

- ✓ Use techniques like Neighborhood Cleaning Rule (NCL) to filter overlapping samples.
- ✓ Apply ensemble models to better handle decision boundary uncertainty.

RECOMMENDATIONS

Integrate Synthetic Data Pipelines into ML Workflows

- ✓ Build end-to-end data pipelines that automate synthetic fraud data generation, validation, and augmentation.
- ✓ Ensure seamless integration with existing ETL, feature engineering, and model training systems.
- ✓ Use version control and lineage tracking to monitor synthetic data evolution and model impact.
- ✓ Incorporate rigorous statistical and domain-expert validation stages to ensure synthetic data quality and relevance.

Periodically Retrain Generative AI Models to Capture Evolving Fraud Patterns

- ✓ Fraudsters continually adapt tactics, so generative models must be retrained on fresh fraud data regularly to reflect new trends.
- ✓ Automate retraining schedules and monitor performance drift with downstream fraud detection models.
- ✓ Leverage domain knowledge and feedback loops from fraud analysts to update generative conditions and rules.

Use Explainability Tools like SHAP to Ensure Fairness and Transparency

- ✓ Apply interpretability frameworks such as SHAP or LIME to both synthetic data and fraud detection models.

- ✓ Detect unintended bias or overfitting to synthetic artifacts.
- ✓ Maintain transparency for regulatory compliance and to build stakeholder trust.
- ✓ Compare feature contributions in real vs synthetic data to identify discrepancies.

Deploy as a Modular Service or API

- ✓ Design synthetic data generation and model training components as modular, composable services or APIs.
- ✓ Enable flexible usage across different teams, projects, or business units.
- ✓ Support scalable deployment in cloud or hybrid environments for resource management.

Implement Rigorous Data Governance and Privacy Safeguards

- ✓ Establish clear policies for synthetic data usage, retention, and access control.
- ✓ Ensure synthetic data does not leak sensitive customer information through privacy audits or differential privacy techniques.
- ✓ Document usage and validation processes for synthetic data comprehensively.

Use Continuous Monitoring and Feedback for Model and Data Quality

- ✓ Monitor fraud detection model performance post-deployment with real-time data.
- ✓ Detect concept drift or unusually high false positive/negative rates.
- ✓ Collect feedback from fraud analysts for continuous improvement.
- ✓ Retrain generative models and detection models dynamically based on this feedback.

Explore Ensemble and Multi-Modal Approaches

- ✓ Combine synthetic data augmentation with other imbalanced data methods like SMOTE or cost-sensitive learning.
- ✓ Use multi-modal synthetic data (tabular + text + images) if applicable for richer fraud scenarios.
- ✓ Leverage ensemble models to capture diverse fraud patterns.

Educate Business and Technical Teams

- ✓ Train teams on the benefits, limitations, and ethical considerations of using synthetic data.
- ✓ Share insights from explainability analyses and validation steps to foster informed decision-making.

CONCLUSION

This project successfully demonstrated the use of Generative AI-powered data augmentation, specifically leveraging CTGAN, for credit card fraud detection. It addressed the significant issue of extreme class imbalance inherent in fraud datasets by generating synthetic minority class samples.

The methodology resulted in a 3% increase in Recall and a 7% improvement in ROC-AUC score on normalized data, highlighting the enhanced detection of fraudulent transactions while minimizing false negatives. These improvements translated into a more reliable and accurate detection system, essential for financial security.

Importantly, the project ensured data compliance and privacy preservation by using synthetic data, thereby avoiding reliance on real sensitive customer data. This approach fosters ethical data practices while allowing dataset expansion as needed to keep pace with evolving fraud patterns.

Overall, the project achieved superior classification performance, confirming the viability of combining CTGAN synthetic augmentation with conventional ML models like Random Forest and Gradient Boosting. These gains showcase the potential of this hybrid approach to strengthen fraud management frameworks in real-world financial applications.

This work paves the way for future research and deployment of synthetic data-driven fraud detection systems, enabling scalable, adaptive, and privacy-conscious solutions to combat increasingly sophisticated financial fraud.

APPENDIX

1. Key Python Libraries Used:

- ✓ pandas, numpy, scikit-learn
- ✓ matplotlib, seaborn
- ✓ CTGAN
- ✓ joblib, XGBoost
- ✓ streamlit

2. Files Included:

- ✓ Project_Presentation.pptx
- ✓ Project_Report.pdf
- ✓ Notebooks/
 - eda_and_preprocessing.ipynb
 - ctgan_and_augmentation.ipynb
 - model_training.ipynb
 - result_visualization.ipynb
- ✓ Datasets/
 - creditcard.csv
 - creditcard_preprocessed.csv
 - creditcard_synthetic.csv
 - creditcard_augmented.csv
- ✓ Model/
 - credit_card_fraud_detection_model.pkl
- ✓ streamlit/
 - app.py
 - requirements.txt
- ✓ README.md

3. Visual Plots for Model Metric Analysis:

